

Max Flow 2

Max-Flow Min-Cut Lecture Outline

1. Max-flow = Min st-cut theorem
 - Image segmentation
2. Correctness of Ford-Fulkerson algorithm

Recap: Ford-Fulkerson

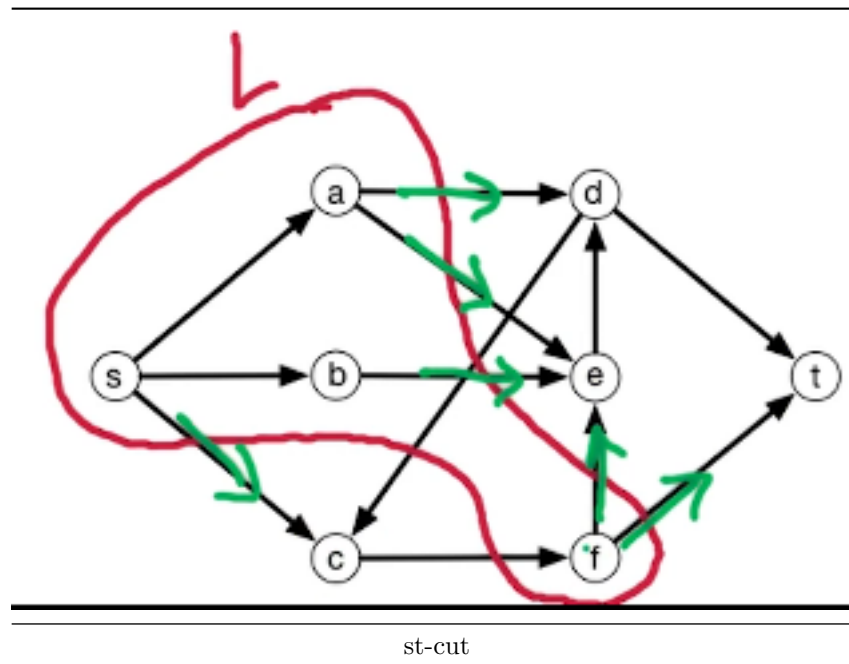
1. Input: Flow network: Directed $G=(V,E)$ with s,t in V and capacities $c_e > 0$
2. Output: flow f^* of maximum size
 - $\text{Size}(f) = f^{\text{out}}(s) = f^{\text{in}}(t)$
3. When does Ford-Fulkerson stop?
 - No augmenting path in residual G^{f^*}
4. Lemma: For a flow f^* if no augmenting path in G^{f^*} then f^* is a max-flow

Quiz: Verifying Max-Flow

1. Given a flow network and a flow f
 - What is the time to check whether or not f is a max flow?
 - $O(n + m)$
 - Build residual graph in $O(n + m)$ time, then check for a path from s to t in this residual graph
 - Simply run DFS to check whether t is reachable from s
 - If there is no path, then f is a max-flow
 - If there is a path, then there's an augmenting path and we can increase the flow

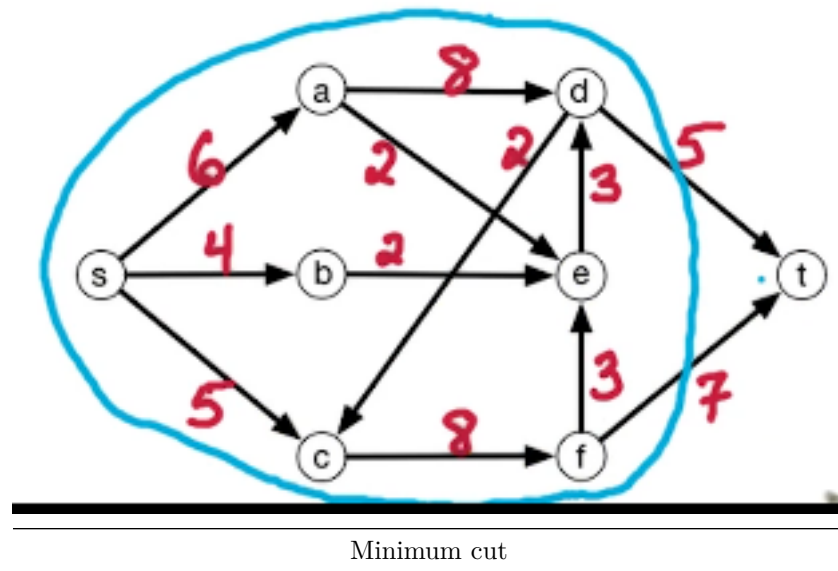
Min-Cut Problem

1. Cut is a partition of $V = L \cup R$
 - st-cut is a cut where s is in L and t is in R
 - Capacity of a cut is sum of capacities of all edges from L to R



Problem Formulation

1. Minimum st-cut problem:
 - Input: Flow network
 - Output: st-cut (L,R) with minimum capacity
2. Max-flow min-cut theorem: Size of max flow is equal to size of min cut



Max-Flow = Min st-Cut

1. Theorem: Size of max-flow = Minimum capacity of a st-cut
 - Creating a minimum cut doesn't necessarily require an s and t like a max-flow does
2. Proof:
 - First, max-flow $\leq$ min st-cut. Then, max-flow $\geq$ min st-cut
 - Proving both proves equality
 - We'll show: For any flow f and any st-cut (L,R)
 - $\text{size}(f) \leq \text{capacity}(L,R)$
 - $\max \text{size}(f) \leq \min \text{capacity}(L,R)$

Max-Flow Min st-Cut

1. We'll show: For any flow f and any st-cut (L,R)
 - $\text{size}(f) \leq \text{capacity}(L,R)$
2. Claim: $\text{size}(f) = f^{\text{out}}(L) - f^{\text{in}}(L)$

Proof of Claim (Max-Flow Min-Cut)

1. Claim: $\text{size}(f) = f^{\text{out}}(L) - f^{\text{in}}(L)$
2. Proof:

Proof:

$$\begin{aligned}
 f^{\text{out}}(L) - f^{\text{in}}(L) &= \sum_{\substack{\vec{vw} \in E: \\ v \in L, w \in R}} f_{vw} - \sum_{\substack{\vec{wv} \in E: \\ w \in R, v \in L}} f_{wv} + \sum_{\substack{\vec{vw} \in E: \\ v \in L, w \in L}} f_{vw} - \sum_{\substack{\vec{wv} \in E: \\ w \in L, v \in L}} f_{wv} \\
 &= \sum_{v \in L} f^{\text{out}}(v) - \sum_{v \in L} f^{\text{in}}(v) \\
 &= \sum_{v \in L-s} (f^{\text{out}}(v) - f^{\text{in}}(v)) + f^{\text{out}}(s) = \text{size}(f)
 \end{aligned}$$

Proof

Finishing Off

1. Claim: $\text{size}(f) = f^{\text{out}}(L) - f^{\text{in}}(L)$
2. Want: $\text{size}(f) \leq \text{capacity}(L, R)$
 - $\text{size}(f) = f^{\text{out}}(L) - f^{\text{in}}(L) \leq f^{\text{out}}(L) \leq \text{cap}(L, R)$
3. Max-flow $\leq$ min st-cut capacity

Reverse Inequality

1. Now we'll show:
 - $\max \text{size}(f) \geq \min \text{capacity}(L, R)$
2. Take flow f^* from Ford-Fulkerson algorithm
 - f^* has no st-path in residual G^{f^*}
 - We'll construct (L, R) where:
 - $\max \text{size}(f) \geq \text{size}(f^*) = \text{capacity}(L, R)$

Proof of Claim (Max-Flow Min-Cut 2)

1. Take flow f^* with no st-path in residual G^{f^*}
 - Let L = vertices reachable from s in G^{f^*}
 - Know t is not in L
 - Let $R = V - L$

Properties of Cut

1. Flow f^* with no st-path in G^{f^*}
 - Capacities of edges aren't important, only which edges are present
 - L = vertices reachable from s in G^{f^*}
 - For $vw \in E$, $v \in L$ and $w \in R$, $f_{vw}^* = c_{vw}$
 - This means the total flow out of L $f^{\text{out}}(L) = \text{capacity}(L, R)$
 - For $zy \in E$, $z \in R$ and $y \in L$, $f_{zy}^* = 0$
 - This means for the total flow into L $f^{\text{in}}(L) = 0$
 - $\text{size}(f^*) = \text{capacity}(L, R)$

Completing the Proof

1. Now we'll show: $\max \text{size}(f) \geq \min \text{capacity}(L, R)$
 - Take flow f^* from Ford-Fulkerson algorithm
 - f^* has no st-path in residual G^{f^*}
 - We'll construct (L, R) where:
 - * $\text{size}(f^*) = \text{capacity}(L, R)$
2. Proves that Ford-Fulkerson stops when there is no augmenting path and outputs the maximum flow
 - Shows we can construct a min st-cut
 - If we take a max flow f^* and set L to be those vertices reachable from s in the residual network, then that st-cut has capacity equal to the size of the flow
 - * Must be a minimum st-cut (optimal)
 - Can both verify a max-flow and create a minimum st-cut